RESEARCH ASSISTANT OPERATIONS MANUAL

Monte Carlo Simulation for Missing Data Imputation

Impact on Regression Robustness

Standardised protocol for evaluating how seven imputation methods preserve core regression conclusions across three missingness mechanisms, seven severity levels, and 30 Monte Carlo iterations.

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Prepared by the Research Team

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1 Research Objective

This manual describes a standardised Monte Carlo simulation framework for evaluating how different missing-data handling strategies affect the robustness of core regression findings in published empirical studies.

Central Research Question

When an empirical study's data contain varying degrees and types of missingness, to what extent do different imputation methods alter the study's core regression conclusions—particularly the sign, magnitude, and statistical significance of its key coefficients?

For each target paper the workflow follows five stages:

1. **Establish the baseline.** Run the paper's main regression on the complete (preprocessed) data to obtain ground-truth estimates.
2. **Introduce missingness.** For each selected key variable, artificially remove values at seven proportions under three mechanisms.
3. **Impute.** Apply seven imputation methods to each incomplete dataset.
4. **Re-estimate.** Re-run the identical regression model on every imputed dataset.
5. **Compare.** Systematically compare the imputed-data regression results against the baseline to quantify distortion.

Every combination is repeated across **30 independent iterations** to yield stable proportion estimates and Wilson confidence intervals.

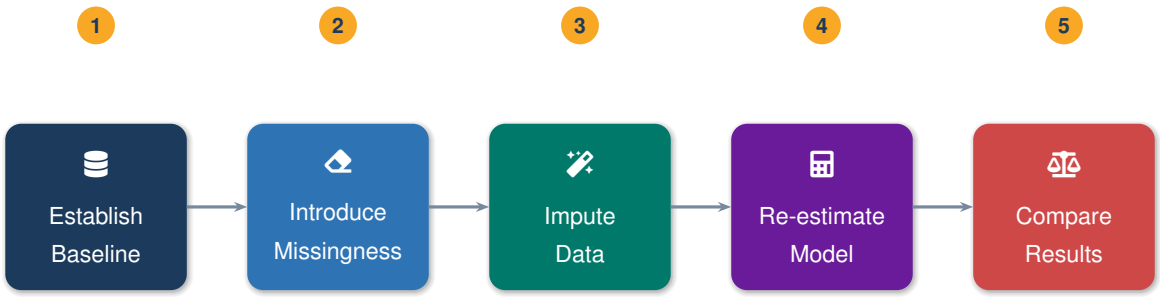

Figure 1: Five-stage pipeline applied to every target paper.

2 Information to Extract from Each Paper

Upon receiving a paper's `.do` file and `.dta` dataset, the RA must identify and document every item below. **This is the single most critical step**—all downstream simulation validity depends on it.

2.1 Data-Structure Information

Item	Description	Example
Data type	Cross-section, panel, repeated cross-section, or other	Panel data
Unit-ID variable	Variable that uniquely identifies each unit	`firm_id`
Time variable	Time dimension variable (if applicable)	`year`
Dataset dimensions	Total rows $\times$ total columns	$10\,000 \times 45$
Existing missingness	Variables already containing missing values and their rates	`income`: 12%

Table 1: Data-structure items to document.

2.2 Core Regression Model Information

Locate the paper's **main regression specification** (typically the most important column in the paper's main results table) from the `.do` file and record the items in Table 2.

Item	Description	Example
Dependent variable	Left-hand-side variable	`ln_gdp`
Core IV / Treatment	Key explanatory variable or interaction term	`Post`$\times$`Treat`
Control variables	All covariates included	`age`, `size`
Regression type	OLS, FE, Logit, IV/2SLS, DID, etc.	Entity-FE OLS
Fixed effects	Which dimensions are absorbed	Entity + Year FE
Clustered SEs	Clustering level	By `firm_id`
Weights	Weight variable (if any)	`pop_weight`
Sample restrictions	Filters applied in original code	`year`$\geq$`2000`
FE-absorbed vars	IVs collinear with FE	`Treatment`

Table 2: Regression-model items to document.

2.3 Selection of Key Variables

Select **3–5 key variables** as targets for artificial missingness introduction. The selection criteria, in descending order of priority, are as follows.

i. **Prefer continuous numerical variables,** because all seven imputation methods are designed primarily for continuous data.

ii. **Prefer substantively important variables** in the original regression: the core independent variable, influential controls, or variables that form interaction terms.

iii. **The variable must have sufficient variability** (not constant or near-constant) and must not be entirely missing in the original data.

iv. **Cover diverse variable types** where possible to test imputation performance across different distributional characteristics.

Variables That Must NOT Be Selected

- The dependent variable itself
- ID variables or time variables
- Binary treatment dummies (e.g., `Treatment`)
- Derived variables fully determined by other variables

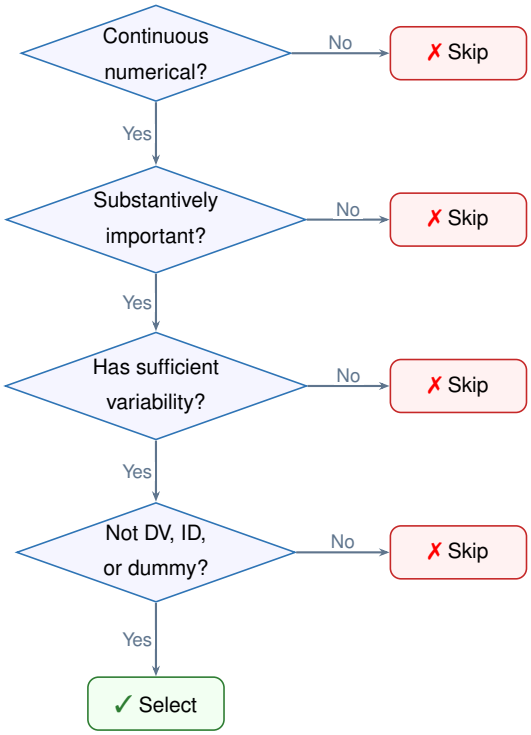

Figure 2: Decision tree for selecting key variables.

2.4 MAR Control Variable

The MAR (Missing at Random) mechanism requires a designated **control variable** whose values determine the probability of missingness. This variable should satisfy all of the following conditions: numerical (continuous or ordinal with many levels), complete or nearly complete, plausibly correlated with most key variables, and *not* one of the key variables selected for missingness introduction. Typical choices include `age`, `firm_size`, `year`, or other widely available variables with ample variation.

3 Simulation Design Parameters

The following parameters are held **constant across all papers** and **do not require per-paper adjustment** unless explicitly approved by the project lead.

3.1 Missingness Mechanisms (3)

Code	Full Name	Meaning	Simulation Method
MCAR	Missing Completely at Random	Missingness is unrelated to any variable	Non-missing values are deleted with equal probability
MAR	Missing at Random	Missingness depends on *other observed* variables	Weighted deletion using the MAR control variable; higher values lead to higher deletion probability
NMAR	Not Missing at Random	Missingness depends on the *variable's own* value	Weighted deletion using the target's own values; higher values lead to higher deletion probability

Table 3: Three missingness mechanisms.

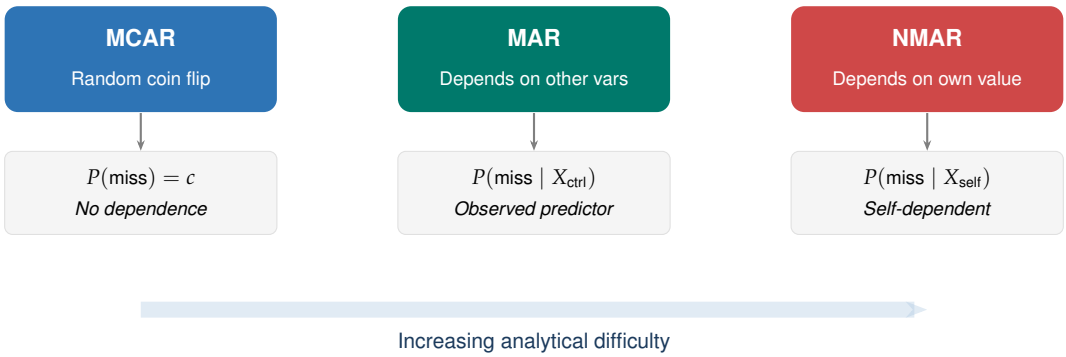

Figure 3: Conceptual comparison of the three missingness mechanisms.

Weighting Implementation

For MAR and NMAR the reference values are normalised and exponentiated with a **strength factor of 1.5**:

$$w_i = \exp(\text{strength} \times \tilde{x}_i), \qquad p_i = \frac{w_i}{\sum_j w_j}$$

where $\tilde{x}_i$ is the normalised value. Observations are then sampled *without replacement* according to $\{p_i\}$.

3.2 Missingness Proportions (7 Levels)

Level	1%	5%	10%	20%	30%	40%	50%
Severity	Negligible	Low	Mild	Moderate	Substantial	High	Very high

Table 4: Seven missingness-proportion levels.

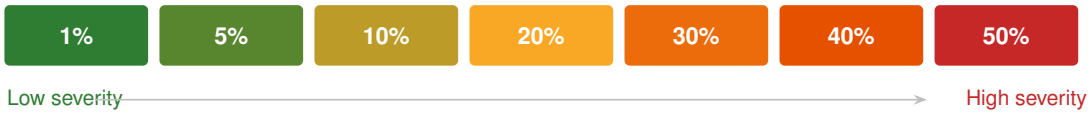

Figure 4: Missingness severity gradient from 1% to 50%.

3.3 Imputation Methods (7)

#	Method	Code	Description
1	Listwise Deletion	LD	Drops all rows with missing values in the target. No actual imputation.
2	Mean Imputation	Mean	Replaces missing values with the arithmetic mean. Underestimates variance.
3	Regression (+ Noise)	Reg	Predicts via linear regression on other variables, then adds residual noise.
4	Stochastic Iterative	Iter	Iterative imputer (Bayesian Ridge); analogous to MICE with posterior sampling.
5	Random Forest (+ Noise)	RF	Predicts via Random Forest, then adds residual noise. Captures non-linearity.
6	Deep Learning (+ Noise)	DL	Predicts via MLP neural network, then adds residual noise.
7	Multiple Imp. (LGBM)	MI	MICE-based MI using LightGBM (M=5); pooled via Rubin's Rules.

Table 5: Seven imputation methods.

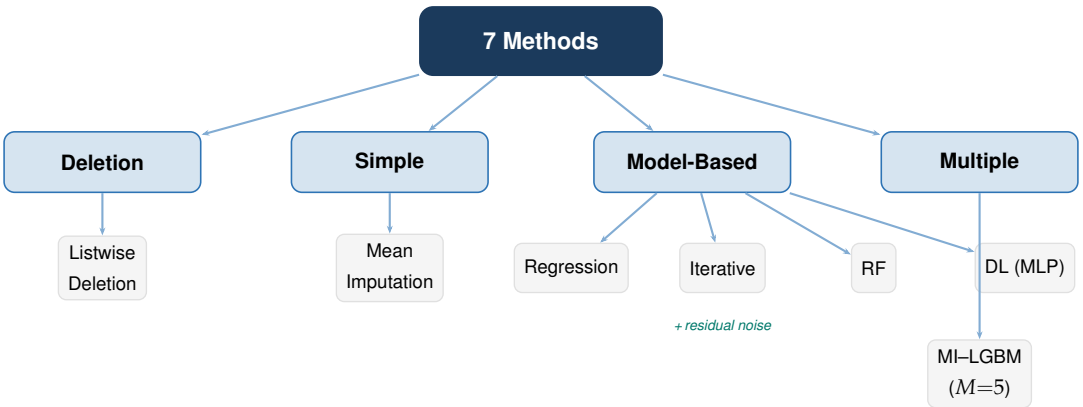

Figure 5: Taxonomy of the seven imputation methods.

On Residual Noise and Multiple Imputation

Noise. Methods 3, 5, and 6 add random noise drawn from the training-set residual distribution to prevent artificially deflated variance.
Multiple Imputation. Method 7 produces M=5 completed datasets (5 MICE iterations each). Regression results are pooled using Rubin's Rules to properly account for imputation uncertainty.

3.4 Number of Iterations

Each unique combination of *mechanism* × *proportion* × *key variable* is repeated over **30 independent iterations**.

3.5 Total Simulation Scale

For a paper with K key variables the total number of imputation-plus-regression runs is:

$$\underbrace{3}_{\text{mech.}} \times \underbrace{7}_{\text{prop.}} \times K \times \underbrace{30}_{\text{iter.}} \times \underbrace{7}_{\text{methods}} = 4{,}410\,K \tag{1}$$

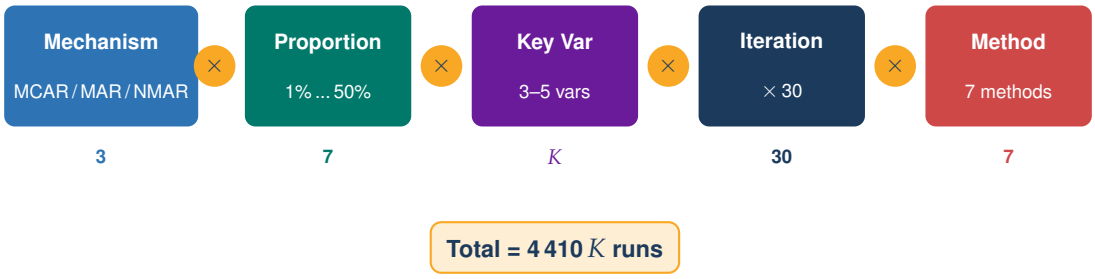

Figure 6: Fully crossed factorial design of the simulation.

4 Step-by-Step Workflow for Each Paper

4.1 Step 1: Data Preparation

1.1. Export the `.dta` file to CSV. In Stata:

```
export delimited using "DATA.csv", replace
```

1.2. Read the `.do` file carefully to understand variable definitions, sample-selection logic, and all variable-generation steps.

1.3. Complete the **Paper Information Record** (Appendix A) with all items from Section 2.

4.2 Step 2: Data Preprocessing Configuration

2.1. Confirm that all numerical variables are correctly typed. Convert categorical variables to dummies or numerical codes.

2.2. Replicate all necessary transforms from the `.do` file: logs, interactions, lags, winsorisation, etc.

2.3. Ensure panel-ID and time variables are correctly set. For panel data, create a composite identifier (e.g. `firmid_year`).

2.4. Apply the original sample restrictions so that the analysis sample matches the original study.

4.3 Step 3: Baseline Regression

3.1. Run the paper's core regression on the complete data.

3.2. Record all coefficients, SEs, p-values, N, and R^2.

3.3. **Validation:** compare against the published table. Minor fourth-decimal-place discrepancies are acceptable.

Baseline Validation Is Mandatory

Do **not** proceed until the baseline output matches the published table. A substantial mismatch invalidates all subsequent simulation results.

4.4 Step 4: Configure Simulation Parameters

4.1. Specify the key-variable list (3–5 variables; see Section 2.3).

4.2. Designate the MAR control variable (Section 2.4).

4.3. Confirm the regression formula, FE, clustered SEs, and weights.

4.4. List all FE-absorbed variables.

4.5. Define the predictor pool for imputation models.

4.5 Step 5: Run the Simulation

5.1. Execute the full simulation pipeline.

5.2. **Test run first:** start with a reduced configuration (2 iterations, 2 levels) to verify correctness.

5.3. Note wall-clock time. Typical runtime ranges from several hours to tens of hours.

4.6 Step 6: Results Inspection and Reporting

6.1. Verify all sheets in the Excel report are populated.

6.2. Inspect for anomalies (all-zero rows, empty rows, implausible values).

6.3. Confirm that every method has output at every missingness level.

6.4. Complete the QC Checklist (Section 6).

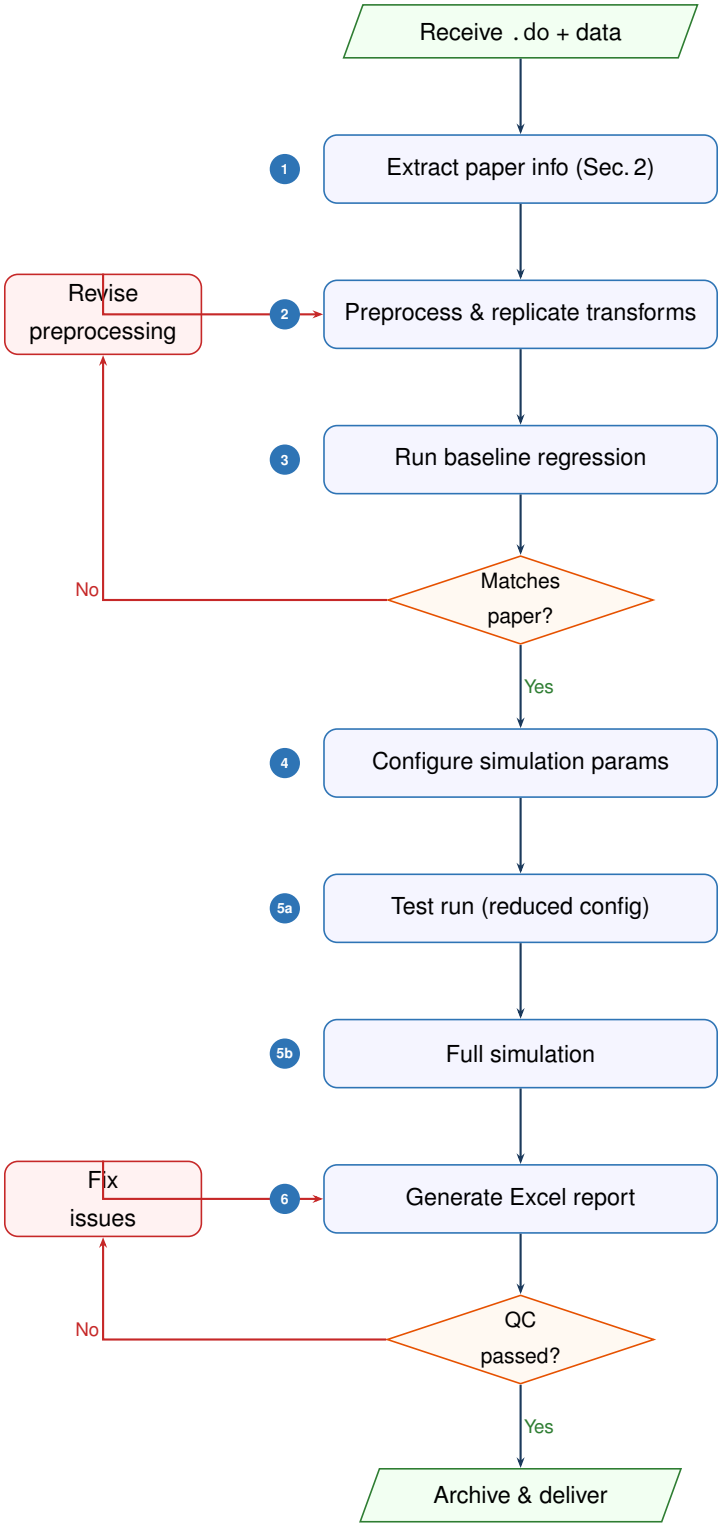

Figure 7: End-to-end workflow for processing a single paper.

5 Interpretation of Output Metrics

5.1 Coefficient Stability (Primary Metric)

For the paper's core coefficient, two proportions are tracked across the 30 iterations:

"Both Same" (B) Proportion

The fraction of iterations in which the imputed-data coefficient simultaneously has the **same sign** *and* the **same significance status** ($\alpha = 0.05$) as the baseline. A higher B-proportion indicates greater robustness.

"Sign Same, Significance Changed" (SS)

The fraction of iterations where the sign is preserved but significance status flips. This captures inference disruption without directional reversal.

Both are reported with **Wilson confidence intervals**.

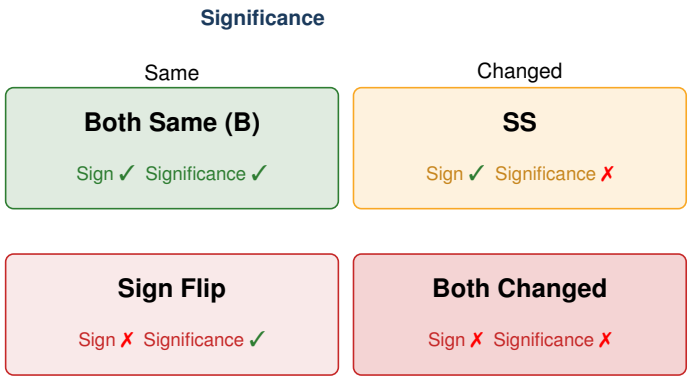

Figure 8: Classification matrix for coefficient stability outcomes.

5.2 Model Comparison Metrics

RMSE

Root-mean-squared deviation between imputed and baseline coefficients, across all common coefficients. Lower is better.

Average Relative Standard Error

Ratio of imputed-data SE to baseline SE, averaged across coefficients. Values above 1 indicate inflated uncertainty; below 1 suggests artificial precision.

Significance and Sign Flips

Indicators of whether any coefficient experienced a significance reversal or a sign reversal.

5.3 Distributional Diagnostics

Variance

Post-imputation variance versus original. Mean imputation is known to deflate variance.

Skewness

Post-imputation skewness versus original.

5.4 Multiple-Imputation Diagnostics (Method 7)

FMI

Fraction of Missing Information: proportion of variance attributable to missingness.

Relative Efficiency (RE)

Efficiency of M=5 imputations. Values above 0.99 are adequate.

MCSE / SE Ratio

Monte Carlo SE divided by pooled SE. If above 0.10, M may be insufficient.

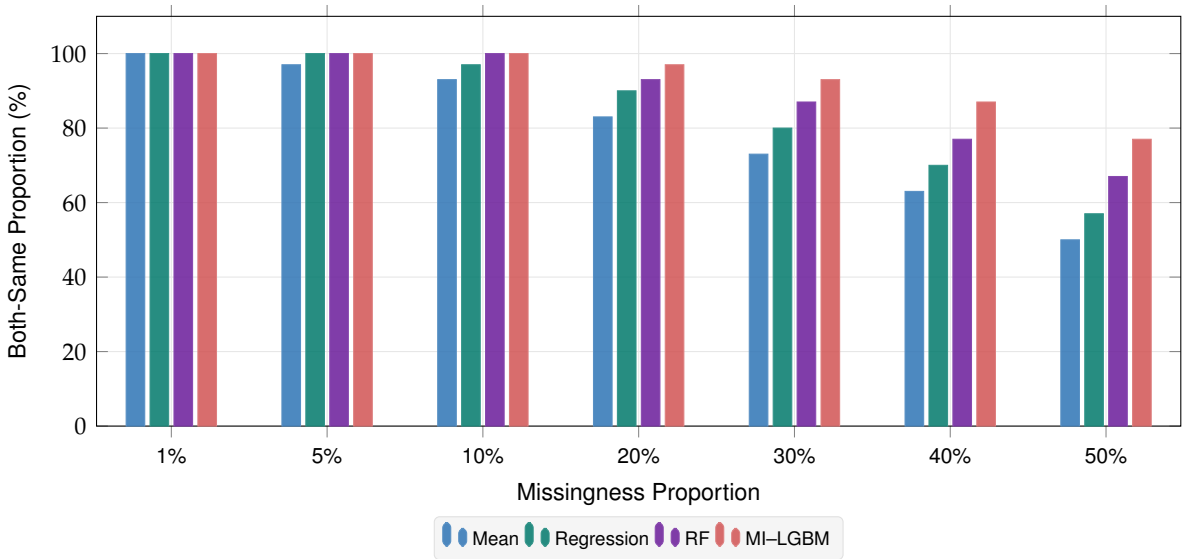

Figure 9: Illustrative "Both Same" proportions under MCAR (hypothetical data for layout reference).

6 Quality-Control Checklist

After completing each paper, verify every item below.

#	Check Item	Pass?
1	Baseline results closely match the published table	☐
2	All 3 mechanisms (MCAR, MAR, NMAR) have been run	☐
3	All 7 proportions (1%–50%) have been run	☐
4	All 7 imputation methods produced output	☐
5	Each combination completed 30 iterations	☐
6	No blank or all-zero sheets in the Excel report	☐
7	At 1% missingness, B-proportion $\approx$ 100%	☐
8	LD at 50% shows substantially reduced N	☐
9	Paper Information Record is complete	☐
10	Files archived per directory convention	☐

Table 6: Quality-control checklist.

7 Frequently Asked Questions

Q1: Which regression model should be selected when multiple are reported?

Select the paper's **preferred specification**—typically the column with all controls and fixed effects. If the paper explicitly states its main model, follow the paper. If multiple are equally central, select 2–3 representative ones and run each separately.

Q2: What if the paper uses IV / 2SLS?

Document this and consult the project lead before proceeding. IV regressions require special treatment.

Q3: Can dummy variables be selected as key variables?

Not recommended. The framework targets continuous variables. If absolutely required, a categorical imputation approach would be needed.

Q4: What if the dataset exceeds 1 million rows?

Consider stratified subsampling while preserving structural integrity. Document the procedure and discuss with the project lead.

Q5: Are 30 iterations sufficient?

In most cases, yes. If specific combinations exhibit very wide CIs, increase to 50 or 100 with documentation.

Q6: Is the MAR/NMAR strength of 1.5 appropriate?

It represents a moderate departure from MCAR. It may be increased to 2.0–2.5 with project-lead approval. **All papers must use the same setting.**

Q7: What if the baseline cannot be exactly replicated?

Fourth-decimal-place differences are acceptable. Deviations $> 5\%$ in the core coefficient require a preprocessing review.

Q8: Should all models be simulated or just the main one?

Default: **main model only**. Alternative specifications may be added as supplementary runs upon request.

8 File Naming and Directory Structure

Each paper's outputs should be stored in an isolated directory following the convention below.

Directory Layout

```
Paper_AuthorYear/
  DATA.csv
  Paper_Info_Record.pdf
  config_notes.txt
```

```
AuthorYear_Report.xlsx
regression_txt_outputs/
  MCAR/
    1pct/ . . .  50pct/
      VarName/
        method_name/
          iter0_model_key.txt
  MAR/
  NMAR/
temp/
```

9 Quick-Reference Summary

Parameter	Value
Missingness mechanisms	MCAR, MAR, NMAR (3)
Missingness proportions	1%, 5%, 10%, 20%, 30%, 40%, 50% (7)
Imputation methods	LD, Mean, Reg, Iter, RF, DL, MI–LGBM (7)
Key variables per paper	3–5 (paper-specific)
Iterations per combination	30
MI datasets (M)	5
MICE iterations	5
Significance threshold	$\alpha = 0.05$
MAR / NMAR strength	1.5
Residual noise	Added for Reg, RF, DL methods
Parallel processing	Enabled (half of available CPU cores)

Table 7: Quick-reference parameter summary.

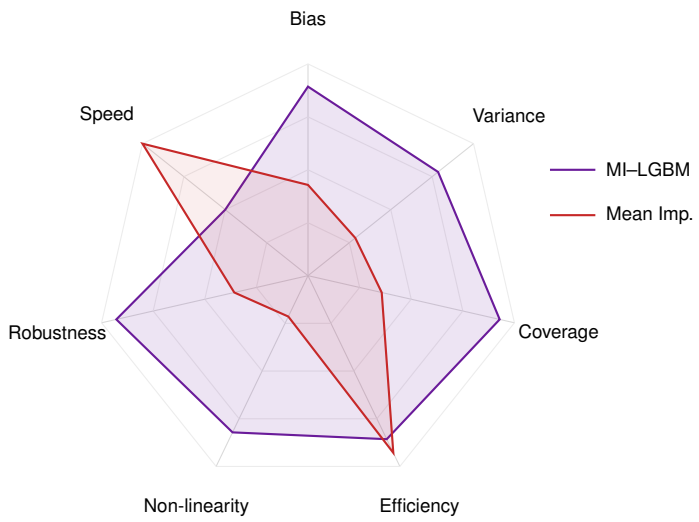

Figure 10: Illustrative radar chart comparing two methods across seven quality dimensions (hypothetical scores).

A Paper Information Record (Template)

Paper Information Record

General Information

Paper ID: .....

Title: .....

Authors: .....

Year & Journal: .....

Data Structure

Data type: ☐ Cross-section ☐ Panel ☐ Other

Unit-ID variable: .....

Time variable: .....

Dimensions: ..... rows × ..... columns

Core Regression Model

Dependent variable: .....

Core IV / interaction: .....

Controls: .....

Regression type: ☐ OLS ☐ FE ☐ Logit ☐ IV ☐ Other

Fixed effects: .....

Clustered SEs: .....

Weights: .....

Sample restrictions: .....

FE-absorbed variables: .....

Simulation Configuration

Key variables: 1. ..... 2. ..... 3. .....

4. ..... 5. .....

MAR control variable: .....

Predictor pool: .....

Baseline Validation

Core coefficient: .....

Core SE: .....

Core p-value: .....

Discrepancy notes: .....

Run Log

Start time: .....

End time: .....

Errors: ☐ No ☐ Yes: .....

QC passed: ☐ Yes ☐ No: .....

B Excel Report Sheet Descriptions

Sheet Name	Contents
`Baseline_Descriptives`	Descriptive statistics for selected variables
`Baseline_Correlations`	Correlation matrix with significance stars
`Baseline_Regression`	Baseline regression output (ground truth)
`Mean_Stability_MCAR`	B and SS proportions with Wilson CIs (MCAR)
`Mean_Stability_MAR`	Same, under MAR
`Mean_Stability_NMAR`	Same, under NMAR
`Model_Comparison`	RMSE, Avg Rel SE, flip indicators
`Stats_Features`	Average variance and skewness of imputed vars
`Coef_Stability_Summary`	Aggregate stability percentages
`Benchmark_*`	Grouped benchmark tables
`MI_Diagnostics`	FMI, RE, MCSE/SE diagnostics
`MI_Trace`	Per-imputation coefficient traces
`MI_Overimputation`	RMSE, MAE, coverage, SMD
`MI_Distribution`	Observed vs. imputed comparisons
`Missingness_Patterns`	Variable-level missing rates
`NMAR_Residual`	Logistic regression of missingness on residuals
`NMAR_Delta`	Sensitivity analysis under delta adjustments

Table 8: Description of Excel report sheets.

C Expected Performance Heatmap (Illustrative)

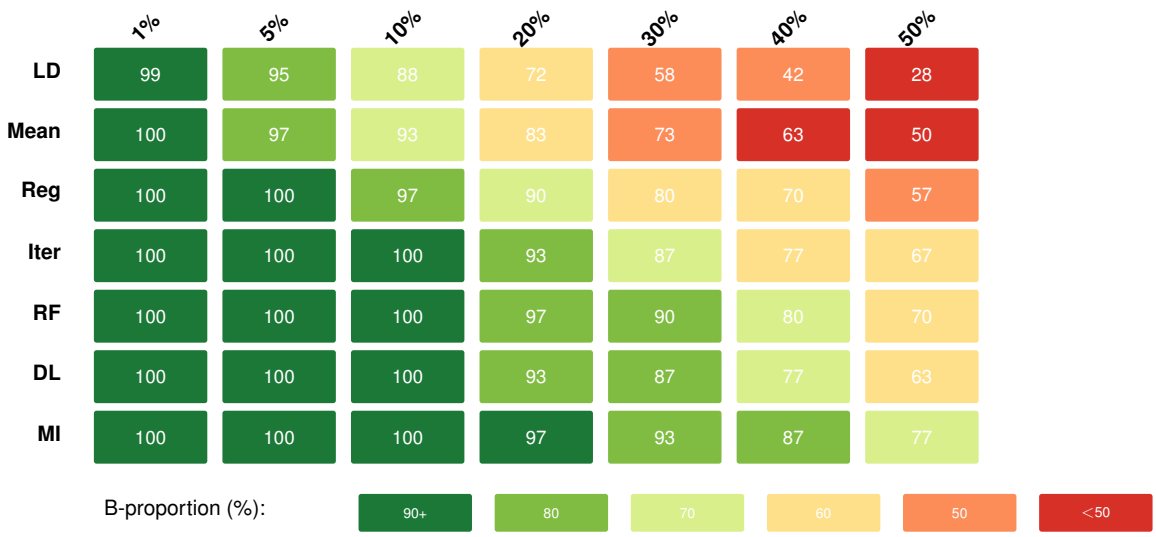

Figure 11: Illustrative heatmap of "Both Same" proportions by method and missingness level under MCAR (hypothetical data for layout reference).

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